

AlgoChowk — Quant Engineer Intern

Quantitative Research & Event-Driven Backtesting Challenge

Time: 6–8 hours

Deadline: 3 days

AI Tools: Allowed & encouraged

Trading experience: Not required

The Challenge

Investigate the hypothesis:

“After a significant one-day fall in NIFTY, the market tends to recover over the next few trading days.”

You are responsible for **sourcing the NIFTY historical daily data** required for the investigation.

Use a credible source and document the **source, date range, fields, and cleaning steps**.

Minimum data:

Date, Open, High, Low, Close

Do not spend excessive time searching for the “perfect” dataset. A credible, consistently documented source is sufficient.

There is **no predefined definition** of “significant fall” or “recovery.” Define them yourself and explain your reasoning.

Treat this as one research investigation, not a checklist.

1. Research Design

Prepare a **Research Note — maximum 2 pages** covering:

- Hypothesis
- Event & recovery definitions
- Entry / exit
- Holding period
- Test period
- Transaction costs
- Key assumptions

Make the hypothesis **precise, measurable and reproducible**.

2. Data & Research Engine

Build a simple **Python notebook or application**.

Validate the Data

Check for:

- Missing / duplicate dates
- Incorrect ordering
- Missing or invalid OHLC values
- Suspicious observations
- Actual date coverage

Document important cleaning or exclusion decisions.

Detect & Analyse Events

For each qualifying event, calculate:

- Event date
- Event-day return
- Entry price
- Forward returns for chosen holding period(s)

Use an entry price that could realistically have been known or executed at that time.

Consider whether closely occurring events should be treated as independent observations.

Include relevant statistics such as:

- Number of observations
- Mean / median

- Win rate
- Distribution / volatility

Explain why your chosen statistics matter.

The experiment should be **configurable** — changing the event threshold or holding period should not require rewriting the research engine.

3. Statistical Evidence

Assess the evidence **for or against** the hypothesis.

Choose appropriate methods, such as:

- Confidence intervals
- Hypothesis testing
- Bootstrap

You do not need to use every method.

Explain:

Why did you choose the method, and what does it actually tell you?

Remember:

Statistical significance does not automatically mean economic or trading significance.

4. Baseline & Robustness

Compare event-period returns with **normal NIFTY forward returns over the same period**.

Then test reasonable changes to:

- Event threshold
- Holding period
- Other important assumptions

Discuss:

- Data snooping

- Multiple testing
- Post-hoc parameter selection
- Overfitting

Do not optimise simply to find the best historical result.

5. Out-of-Sample Validation

Split the data into:

Research / Development → Out-of-Sample

Use the first period for research and keep the second period **unseen during development**.

Explain:

- Your split
 - Number of observations
 - Whether the relationship persisted
 - What changed
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6. Challenge Your Own Result

Try to **disprove your conclusion**.

Consider the risks most relevant to your experiment, such as:

- Look-ahead bias
- Slippage / transaction costs
- Sample size
- Overlapping events
- Market regimes
- Data quality

Answer:

What evidence would make you reject your original hypothesis?

7. Simple Event-Driven Backtest

If the research supports a tradable interpretation, implement a simple event-driven backtest.

Include:

- Entry / exit
- Costs and slippage
- Number of trades
- Cumulative performance
- Maximum drawdown

This is an **extension of the research**, not the primary objective.

Do not build a sophisticated backtesting framework. Research quality matters more than return.

8. Conclusion

Summarise:

Hypothesis → Method → Evidence → Baseline → Robustness → Out-of-Sample → Limitations → Conclusion

There is **no expected answer**. A negative result is completely acceptable.

Technology

Preferred:

- Python
- Pandas
- NumPy
- SciPy / statsmodels
- Jupyter
- Git / GitHub

You may research techniques you don't already know.

We care about your ability to **understand and correctly apply** them.

AI Usage

AI tools are **allowed and encouraged**.

You may use ChatGPT, Claude, Gemini, Cursor, Copilot or similar tools.

Include a **1-page AI Usage Note** covering:

- Tools used
- How you used them
- Your own decisions
- Suggestions you changed or disagreed with
- Incorrect AI suggestions you identified
- What you learned

AI can help write the code. It cannot replace your reasoning.

Evaluation

Area	Weight
Quantitative Research & Experimental Design	25%
Statistical Reasoning	20%
Critical Thinking & Falsification	20%
Backtesting Methodology	15%
Engineering / Code Quality	15%
Communication	5%

We are **not primarily looking for**:

- The highest return
- Complex mathematics
- A huge codebase
- A sophisticated strategy

- An elaborate UI

We are looking for someone who can:

Define precisely → Test carefully → Question the result → Validate independently → Explain clearly

Submission

1. **GitHub repository**
2. **Research Note** — max 2 pages
3. **README** — methodology, assumptions, setup, results and limitations
4. **AI Usage Note** — max 1 page
5. **2–3 minute video** showing your approach, findings and key learnings

Your repository should allow a reviewer to **reproduce your main results using your documented data and methodology**.

Final Thought

Don't optimise for a good backtest. Investigate whether the evidence deserves to be believed.